



ACTIVITY IN IAS

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SECTION: BSIT 3-1

ACTIVITY: Comparison

Explain and give simple examples:

Symmetric vs. Asymmetric vs. Hashing vs. Signatures

1. Symmetric Encryption

- Symmetric encryption is a type of cryptography where **only one key is used** to both encrypt and decrypt data.
- Both sender and the receiver must **share the same secret key** before they can communicate securely.
- This method is **fast and efficient**, but it can be risky if the shared key is exposed.
 - Example:
 - You save a document on your laptop and protect it with a password. When you lock the file, the system uses that password to **encrypt** the document. When you want to open it again, you enter the same password to **decrypt** it. Anyone who knows the password can also access the file.

2. Asymmetric Encryption



- Asymmetric encryption is a type of cryptography that uses two different keys: a public key and a private key.
- The public key is used to encrypt data, while the private key is used to decrypt it.
- This makes communication more secure because only the owner of the private key can access the message
 - Example:
 - when you enter your credit card details on a shopping website, your browser uses the website's public key to **encrypt** the information before sending it. Only the website's server, which has the private key, can **decrypt** and read your payment details securely.

3. Hashing

- Hashing is a process that converts data into a fixed-length value called a hash.
- It is a one-way process, meaning the original data cannot be retrieved from the hash.
- Even a small change in the input will produce a completely different hash value.
 - Example:
 - When you register your fingerprint, the system converts it into a hash value and stores it. When you try to unlock your phone, your fingerprint is scanned again and hashed, then compared to the stored hash to check if it matches.

4. Digital Signatures

- Digital signatures are used to **verify the identity of the sender and ensure data integrity**.
- It uses **asymmetric encryption**, where the sender signs using a private key and the receiver verifies using a public key.
- This ensures that the message is authentic and has not been altered.
 - Example:
 - when a company sends you an official email, it includes a digital signature created using its private key. Your email system can use the company's public key to verify that the message really came from them and was not changed during transmission.



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Comparison:

Symmetric	Asymmetric	Hashing	Signatures
One shared key	Two keys (public & private)	No key (algorithm only)	Two keys (public & private) with hashing
Encrypt and decrypt data	Secure communication	Protect data	Verify identity and authenticity
Fast	Slower	Very fast	Moderate
Less secure (key sharing risk)	More secure	Very secure for storage	Very secure for authentication